

PYTHON SET PROGRAMS WITH OUTPUT

Program 1

```
# Set of integers
my_set = {1, 2, 3}
print(my_set)
```

Output:

{1, 2, 3}

Program 2

```
# Set of mixed datatypes
my_set = {1.0, "Hello", (1, 2, 3)}
print(my_set)
```

Output:

{1.0, 'Hello', (1, 2, 3)}

Program 3

```
my_set = {1, 2, 3}
my_set.add(4)
print(my_set)
```

Output:

{1, 2, 3, 4}

Program 4

```
my_set = {1, 2, 3}
my_set.remove(2)
print(my_set)
```

Output:

{1, 3}

Program 5

```
numbers = [1, 2, 2, 3, 4]
my_set = set(numbers)
print(my_set)
```

Output:

{1, 2, 3, 4}

Program 6

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 7, 8}
```

```
distinct_nums = nums1.union(nums2)
```

```
print("The union of two sets is: ", distinct_nums)
```

Output:

The union of two sets is: {1, 2, 3, 4, 5, 6, 7, 8}

Program 7

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 7, 8}
```

```
common_nums = nums1.intersection(nums2)
```

```
print("The intersection of two sets is:", common_nums)
```

Output:

The intersection of two sets is: {4, 5}

Program 8

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 8, 8}
```

```
nums3 = nums1.difference(nums2)
nums4 = nums2.difference(nums1)
```

```
print("nums1 - nums2: ", nums3)
print("nums2 - nums1: ", nums4)
```

Output:

nums1 - nums2: {1, 2, 3}

nums2 - nums1: {6, 7, 8}

Program 9

A = {1, 2, 3, 4}

B = {3, 4, 5, 6}

print(A | B) # Union

print(A & B) # Intersection

print(A - B) # Difference

Output:

{1, 2, 3, 4, 5, 6}

{3, 4}

{1, 2}